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Residual transfusion risk of occult hepatitis B transmission associated to genotypes A and E among blood donors in Yaoundé, Cameroon

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Abstract

Background Occult hepatitis B virus infection (OBI) poses a significant challenge to blood transfusion safety, as it occurs in HBsAg-negative blood donors who are anti-HBc positive and carry detectable hepatitis B virus (HBV) DNA. The lack of additional serological screening and accessible molecular testing in blood banks allows blood units with anti-HBc positivity and HBV DNA to be transfused. The aim of the study was to determine the residual transfusion risk and to identify OBI genotypes among blood donors at the Yaoundé Central Hospital.

Methods This cross-sectional, analytical, and descriptive study involved 269 blood donors between February and June 2025 at the Yaoundé Central Hospital blood bank and at the Centre Pasteur du Cameroun. Informed consent was obtained from all participants. HBsAg was screened through both Rapid Diagnostics Test (RDTs) and Enzyme Linked Immunosorbent Assay (ELISA). ELISA was used for the detection of Anti-HBc IgM and IgG antibodies. Testing for occult HBV infection in the HBsAg-negative, anti-HBc-positive subjects was performed using quantitative polymerase chain reaction (qPCR). HBV genotypes were determined by sequencing a fragment of the surface gene (PreS1) and then performing phylogenetic analysis of the nucleotide sequences. Statistical analysis was performed using R software 4.4.3.

Results Of the 269 donors, 168 (62.45%) were HBsAg-negative and anti-HBc-positive. HBV DNA was detected in 9 donors, confirming occult HBV infection with a residual transfusion risk of 3.34%. The number of prior donations was significantly factor negatively associated with HBV DNA presence ($p = 0.021$, OR [95% CI]: 0.13 [0.0–0.80]). Sequencing of the PreS1 gene fragment from 3 isolates revealed co-circulation of HBV genotypes A and E in the studied population.

Conclusion This study highlights that only sensitive HBV DNA screening would improve blood safety since anti-HBc screening would exclude 60% of donors in a situation of blood shortage. However, the cost of such testing, even in the low yield pools of 6 as practice in some countries, is likely prohibitive in Cameroon. In present Cameroon situation, systematic, compulsory, vaccination would be the best affordable prevention of OBI transfusion transmission.

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Trial registration The study protocol was reviewed and approved by the Regional Ethics Committee for Human Health Research of the Center (CRERSH/C) (Number 000734/ CRERSHC/2025) and adhered to the ethical guidelines outlined in the 1975 Declaration of Helsinki.

Keywords Hepatitis B virus, Occult hepatitis B, Anti-HBc, HBV DNA, Genotype, Blood donors

Introduction

Hepatitis B virus (HBV), an enveloped, circular DNA hepatotropic virus of the Hepadnaviridae family [1], caused an estimated 254 million chronic infections and 1.1 million deaths in 2022, primarily due to cirrhosis or hepatocellular carcinoma (HCC) [2], and when liver failures results from extensive destructive livers cells, therapeutics management becomes very difficult [3]. Screening for HBV infection in pregnant women is essential. In low- and middle-resource countries, hepatitis B endemicity is mainly due to mother-to-child transmission [4]. Other factors of HBV transmission include intravenous drug abuse, occupational exposure to infected blood products, multiple sexual partners, and lack of awareness [5, 6].

HBV is a small virus measuring 42 nm in diameter and consisting of 3,2 kb [7], which encodes four open reading frames (ORF) designated as polymerase (ORF P), precore/core (ORF PreC/C), X (ORF X) and PreS/S (ORF preS/S) [8]. HBV is classified into ten genotypes (A–J) distributed globally [9, 10]. In Cameroon, genotypes A and E co-circulate in comparable proportions, with significant heterogeneity in genotype A (predominantly sub-genotype A3) and homogeneity in genotype E [11].

The WHO goal of eliminating hepatitis B by 2030 appears very difficult to achieve due to phases like the window period, immune control, and occult HBV infection not being detected by standard tests. Sub-Saharan Africa, including Cameroon, is a high-endemicity region with a seroprevalence exceeding 8% [2], but the majority of diagnostic laboratories still use hepatitis B surface antigen (HBsAg)-based tests [12]. The limit of detection (LoD) for the One Step Hepatitis B Surface Antigen Rapid Test (Qingdao Hightop Biotech, China) is typically 0.5–1.0 ng/mL, while the Fortress Diagnostics HBsAg ELISA kit is 0.5-1.0 IU/mL. These thresholds are relatively high compared to modern highly sensitive tests like ARCHITECT (with extremely reduced LoD of 0.005 IU/mL) [13] this is why cases with very low HBsAg concentrations and OBI could be often missed and have not being detected by standard ELISA and RDT HBsAg tests used in Cameroon. Nucleic acid testing for HBV is widely used in developed countries, but its use is limited in low-income countries due to a higher cost and requirement of trained manpower [14].

Occult HBV infection (OBI) is characterized by detectable HBV DNA and serological profile including predominantly anti-HBc positivity but also anti-HBs only, in HBsAg-negative individuals though not accounted in this study. This condition, often undetectable by standard tests, was first hypothesized to involve HBV presence in the liver without circulating HBsAg [15]. International consensus meetings in 2008 and 2018 identified covalently closed circular DNA (cccDNA) as the molecular basis of OBI, resulting from silent mutations in the surface gene open reading frame S (ORF S) specifically in the PreS1, PreS2, and S regions encoding HBsAg [16, 17]. In Cameroon, with an HBV seroprevalence of 11.2% [18]. Despite the high seroprevalence of hepatitis B, only a limited number of studies on occult hepatitis B have been undertaken in Cameroon [19–21]. The aim of the study was to determine the residual transfusion risk and to identify OBI genotypes among blood donors at the Yaoundé Central Hospital.

Materials and methods

Study design, study site and sampling

We conducted a cross-sectional, analytical, and descriptive study from February to June 2025 at the blood bank of the Yaoundé Central Hospital and the virology department of the Centre Pasteur du Cameroun. We included participants over the age of 18 who provided written informed consent and tested negative for HBV, HIV, HCV, and TPHA.

The minimum size for final sample, for identifying OBI among donor was calculated using the Lorentz formula:

$$\mathbf{N} = \frac{\boldsymbol{Z^2} * \boldsymbol{P} * (\mathbf{1} - \boldsymbol{P})}{\boldsymbol{A^2}}$$

Where:

N = minimum sample size
Z = confidence level (1.96 for 95%)
P = The prevalence of OBI in Yaoundé (*P* = 1.55%) [19]
A = margin of error (0.05)

This yielded a minimum of 23 donors.

The minimum sample size was 23 donors, but 269 samples were collected at the prime level for serological testing and selection, hence for statistical robustness.

A 5 mL blood sample was drawn from the antecubital vein into anticoagulant-free vacuum tube. Samples were centrifuged at 5000 rpm for 5 min to obtain serum. Part of the serum was used for rapid diagnostic tests, while the

remainder was aliquoted into 1.5 mL cryotubes, stored at 2–8 °C for serological assays, and frozen at – 80 °C for molecular analyses.

Serological testing

All samples were analyzed following the manufacturer's strict recommendations. Initially, an immunochromatographic rapid test was performed to detect HBsAg using 50 µL serum (One Step Hepatitis B Surface Antigen Rapid Test, Qingdao Hightop Biotech, China). HBsAg was confirmed by Enzyme Linked Immunosorbent Assay (ELISA) 50 µL serum, (Fortress Diagnostics Limited, UK) using a spectrophotometer (SinoThinker, China) at 450 nm after adding stop solution (sulfuric acid). HBsAg-negative samples were tested for anti-HBc (IgM and IgG) by ELISA 50 µL serum, (Fortress Diagnostics Limited, UK) using the same spectrophotometer.

HBV DNA qPCR amplification

Total Nucleic acids were extracted from 250 µL of negative HBsAg and positive IgM or IgG anti-HBc serums using an automated system (Extraction Kit VER 1.0, Hain Lifesciences, Germany) according to the manufacturer's instructions. Quantification of HBV viral load was performed using the Generic HBV-Charge Viral v2.0 kit (Biocentric, France). The reaction mixture was prepared according to the number of PCR reactions. It comprised 10 µl of donor samples, four HBV standards (10 µl each; 5×10^6, 5×10^5, 5×10^4, 5×10^3), 10 µl of the positive control, and a negative control. Additionally, 4 µl of the 5X enzyme mix, 3 µl of the primer-probe mix, and 3 µl of nuclease-free water were thawed, and the final reaction mixture was assembled in a 1.5 mL nuclease-free tube. Amplification was performed in a QuantStudio™ 5 Real-Time PCR system using the following program: 50 °C for 15 min,95 °C for 10 min,95 °C for 15 s and 60 °C for 1 min. Viral load values are considered positive with a load ≥ 11 IU/mL (1.04 log IU/mL).

HBV PreS1 region sequencing

Using the DNA extracts obtained after amplification, we did not succeed, so we decided to use a manual extraction for the remainder of the work. Manual DNA extraction from 200 µL serum (Invitrogen Kit, France) according to the manufacturer's instructions. Pre-S region for HBV genome was amplified by nested PCR using two sets of primers as previously described with modifications [22]. Briefly, the first PCR was performed in a final volume of 50 µL containing 1U of Taq polymerase (Life Technologies Corporation, USA), 5 µL of 10x buffer, 2 µL of MgCl2, 2 µL of dNTPs, 2 µL of primers HBPr1 (nt 2850–2868, 5'-GGGTCACCATATTCTTGGG-3') and HBPr135 (nt 803–822, 5'-CAAGACAAAAGAAAATTG G-3'), 10 µL of DNA and 26 µL of Rnase free water. For the second step, 4 µL of PCR products were used in a final volume of 50 µL containing 1U of Taq polymerase (Life Technologies Corporation, USA), 5 µL of 10x buffer, 1 µL of MgCl2, 2 µL of dNTPs, 2 µL of primer HBPr2 (nt 2867–2888, 5'-GAACAAGAGCTACAGCATGGG-3') and HBPr3 (nt 1547–1569, 5'-CCACTGCATGGCCTGA GGATG-3') and 33 µL of Rnase free water. The first and second round PCR conditions were one cycle at 95 °C for 5 min, followed by 35 cycles at 95 °C for 30 s, 50 °C for 30 s, and 72 °C for 30 s, with a final extension of 72 °C for 7 min.

Amplification was performed in a SimpliAmp Thermal Cycler (Applied Biosystems™, USA). Throughout the sample analysis process, from DNA extraction to sequencing of nested PCR amplicons, a positive control (sample with a known positive result) and negative control (Rnase free water) were used to validate or invalidate the analysis process.

Phylogenetic analysis

Nested PCR products were visualized after electrophoresis on a 2% agarose gel. Sanger bidirectional sequencing (forward and reverse) was performed using the BigDye Terminator technology (Inqaba Biotec, Pretoria, South Africa). Consensus sequences were obtained after manual editing of forward and reverse sequences using CLC Main Workbench software (Version 5.5.2). Reference HBV sequences were obtained from GenBank. Phylogenetic trees were constructed using the neighbor-joining method in MEGA 7.0, with genetic distances calculated via the Kimura-2-parameter model. Bootstrap analysis (1000 replicates) displayed values > 70% on tree branches. Nucleotide sequences of identified HBV genotypes were deposited in GenBank (www.ncbi.nlm.nih.gov).

Statistical analysis

Sociodemographic, clinical, and biological data were compiled in Excel 2021 and analyzed using R software 4.4.3. Results are presented as frequencies, means (± standard deviation), ratios, and proportions. We used univariate analysis to evaluate associations using Fisher test, with a significance threshold of $p = 0.05$. Odds ratios (OR) and 95% confidence intervals (CI) assessed the strength of associations between outcomes and independent variables.

Results

Sociodemographic and clinico-biological characteristics

Of the 269 participants, 231 (86%) were male, and 38 (14%) were female, yielding a male-to-female ratio of 6.08. The mean age was 30.65 ± 8.21 years. Single individuals predominated (205, 79.5%), followed by married individuals (61, 22.7%). Donors with higher education (126, 48.1%) and secondary education (114, 43.5%) were most

represented. Family donors comprised 82.16% (221/269) and 5.20% (14/269) for volunteer donors12.7% (34/268) of paid donors, with 53.7% being first-time donors, and 96.3% had never received a transfusion (Table 1).

Anti-HBc seroprevalence

Anti-HBc seroprevalence was 62.45% (168/269), with males (143/168) more affected than females (25/168). Age was significantly associated with anti-HBc presence ($p < 0.001$), with 91.30% (63/69) in the 40–50 + age group versus 47.50% (57/120) in the 18–29 group (Table 1).

Table 1 Association between donor characteristics and anti-HBc prevalence

Characteristic	HBcAb		*P*–value
	Presence n (%)	Absence n (%)	
Sex			0.720
Male	143 (61.1)	88 (38.1)	
Female	25 (65.8)	13 (34.2)	
Profession			0.520
artisan	2 (100.0)	0 (0.0)	
Arts and leisure	2 (28.6)	5 (71.4)	
student	66 (58.9)	46 (41.1)	
Public Service	20 (69.0)	9 (31.0)	
Agriculture	5 (71.4)	2 (28.6)	
Service Industry	22 (59.5)	15 (40.5)	
Technical professions	16 (57.1)	12 (42.9)	
Self-Employed	13 (72.2)	5 (27.8)	
Age Group			**< 0.001**
[18–29]	57 (47.5)	63 (52.5)	
[30–39]	31 (49.2)	32 (50.8)	
[40–56]	63 (91.1)	6 (8.7)	
Marital Status			0.525
Single	130 (63.4)	75 (36.6)	
Married	36 (59.0)	25 (41.0)	
Divorced/Widowed	2 (100.0)	0 (0.0)	
Education level			0.246
Primary	17 (77.3)	5 (22.7)	
Secondary	73 (64.0)	41 (36.0)	
Higher	74 (58.7))	52 (41.3)	
Prior Donation			0.801
0	91 (63.2)	53 (36.8)	
1+	76 (61.3)	48 (38.7)	
Prior Transfusion			0.32
Yes	8 (80.0)	2 (20.0)	
No	159 (61.6)	99 (38.4)	
Donation Type			0.966
Family	137 (62.0)	84 (38.0)	
Volunteer	9(64.3)	5(35.7)	
Paid	22 (64.7)	12 (35.5)	

Occult hepatitis B virus infection prevalence and associated risk factors

Overall, OBI prevalence was 3.34% (9/269), with males (6/9) predominating (sex ratio: 2). Single donors (7/9) and those with secondary education (5/9) were most represented. The number of prior donations was significantly associated with HBV DNA presence ($p = 0.021$), with first-time donors overrepresented (8/9) (Table 2).

All 9 OBI donors were family donors, aged 22–48 years, with viral loads ranging from < 11 IU/mL (lower limit of quantification, LLoQ) to 9437.234 IU/mL (highest in the youngest donor) (Table 3).

HBV genotype identification in blood donors with OBI

Of the three sequences obtained (HBO131, HB079, HB084), HBO131 belonged to genotype A, and HB079 and HB084 belonged to genotype E. These two genotypes are prevalent in sub-Saharan Africa, where they co-circulate in certain countries such as Cameroon. Hence their strong similarity with sequences from these countries (Fig. 1).

The HBV strains identified in this study are available in GenBank (GenBank accession numbers: PX313721-PX313723).

Discussion

Hepatitis B virus remains a transfusion safety concern, particularly in resource-limited settings where pre-transfusion testing focuses on HBsAg, missing OBI cases [23]. The aim of the study was to determine the residual transfusion risk and to identify OBI genotypes among blood donors at the Yaoundé Central Hospital.

The study population was predominantly male (86%, sex ratio: 6), possibly due to physiological factors (e.g., pregnancy, breastfeeding) or psychological barriers (e.g., fear of needles) limiting female donors. This aligns with a 2013 study in Cameroon reporting a sex ratio of 4.5 [24]. The mean weight (78.85 ± 13.29 kg) met WHO minimum donor weight of 50 kg [25], and the mean age (30.65 ± 8.21 years) complied with Cameroon National Blood Transfusion Program regulations. These findings are consistent with studies in Somalia (31.5 ± 8.11 years) [26] and Nigeria (31.7 ± 7.9 years) [27].

OBI is characterized by the presence of HBV DNA in low concentrations in the serum (< 200 IU/mL) of a patient who tests negative for hepatitis B surface antigen (HBsAg) [28]. Two scenarios often confused with OBI include the seroconversion phase, where HBsAg is absent while cccDNA is detectable in hepatocytes, and the resolution phase, where HBsAg clears before cccDNA, which persists temporarily [12]. Anti-HBc is a marker of prior HBV exposure and a potential indicator of OBI in blood donors. In high-endemicity regions, anti-HBc seroprevalence is elevated [29], and the risk of HBV transmission

Table 2 OBI prevalence and associated risk factors

Characteristics	*n* (%)	HBV DNA		*P*–value
		Detectable (%)	OR (95% CI)	
Sex				0.137
Male	231 (86.5)	6 (2.6)	1	
Female	38 (13.5)	3 (7.9)	3.2 [0.65–12.8]	
Age group				0.1263
[18–29]	143 (53.8)	3 (2.1)	1	
[30–39]	89 (33.1)	3 (3.4)	1.6 [0.29–8.97]	
[40–56]	37 (13.1)	3 (8.1)	4.1 [0.73–23.1]	
Marital statues				0.092
Single	205 (76.4)	7 (3.4)	1	
Married	61 (23.2)	1 (1.6)	0.5 [0.0–2.7]	
Divorced/Widowed	2 (0.4)	1 (50.0)	28.3 [1.0–767.8]	
Education level				0.185
Primary	22 (7.9)	2 (9.1)	1	
Secondary	114 (43.1)	5 (4.4)	0.5 [0.1–3.6]	
Higher	126 (49.0)	2 (1.6)	0.2 [0.0–1.4]	
Prior donations				**0.021**
0	144 (52.5)	8 (5.6)	1	
1+	124 (11.1)	1 (0.8)	0.13 [0.0–0.80]	
Prior transfusion				
No	258 (96.1)	9 (3.5)		
Yes	10 (3.9)	0 (0.0)	NA	
Donation Type				
Family	225 (86.9)	9 (3.8)		
Paid	34 (13.1)	0 (0.0)	NA	

The above table shows a significant association between the number of prior donations and viral load ($p = 0.021$). First-time donors had a 5.6% HBV DNA positivity rate compared to 0.8% for repeat donors. The odds ratio (OR) of 0.138 suggests that repeat donors have an 86% lower chance of HBV DNA positivity. NA: Not applicable

Table 3 Characteristics of HBV DNA-positive donors

Code	Age	Sex	Viral load (IU/mL)
HB030	40	F	12.528
HB047	42	M	34.91
HB079	48	F	21.014
HB084	30	M	< 11*(LLoQ)*[*]
HBO131	28	M	25.687
HBO172	37	M	142.392
HBO222	22	F	9437.234
HBO243	25	M	352.433
HBO281	39	M	5354.835

[*]LLoQ: lower limit of quantification; This sample showed a clear sigmoidal curve in qPCR despite the fact that the load was below the kit's limit of quantification

through transfusion from anti-HBc-positive, HBsAg-negative donors was first noted in 1978, highlighting the concept of OBI [30].

We found anti-HBc seroprevalence of 62.45% among HBsAg-negative donors, indicating significant prior HBV exposure. This rate is higher than the 57% reported at Laquintinie Hospital, Douala [12] but lower than 75.74% found previously at the Yaoundé Central Hospital [31]. A recent study reported 58% at Buea's Regional Blood Bank [21]. In developed countries, systematic anti-HBc screening minimizes residual HBV transmission risk [32]. In contrast, Egypt reported lower anti-HBc prevalence of 7.8% [33] and 16.6% [34], reflecting global variations in HBV seroprevalence,

OBI results from silent mutations enabling HBV DNA replication in the liver without detectable HBsAg, with low cccDNA replication (< 200 IU/mL) [35]. It can progress to chronic phases or HCC [36] and is transmissible via transfusion. OBI prevalence depends on the population studied, diagnostic techniques, and HBV endemicity [37].

This study found a 3.34% OBI prevalence among blood donors, higher than 1.55% previously reported [19] at the same hospital, possibly due to using ELISA versus immunochromatographic tests for anti-HBc and HBsAg, and qPCR versus conventional PCR for viral load. Our results align closely with Mbencho et al., 2024 at 4.5%, using similar methods [21]. In Nigeria, a higher (17%) OBI prevalence was reported [38], likely due to a larger sample (n = 429) and targeting the full HBsAg-coding genome, In 2013 Said Zn et al.. determined a similarly very high prevalence in blood donors (17.2%) [34]. In Pointe Noire(Republic Of Congo) a study conducted by Angounda et al. revealed a prevalence of 2.3% among blood donors [39] which is slightly lower than our results

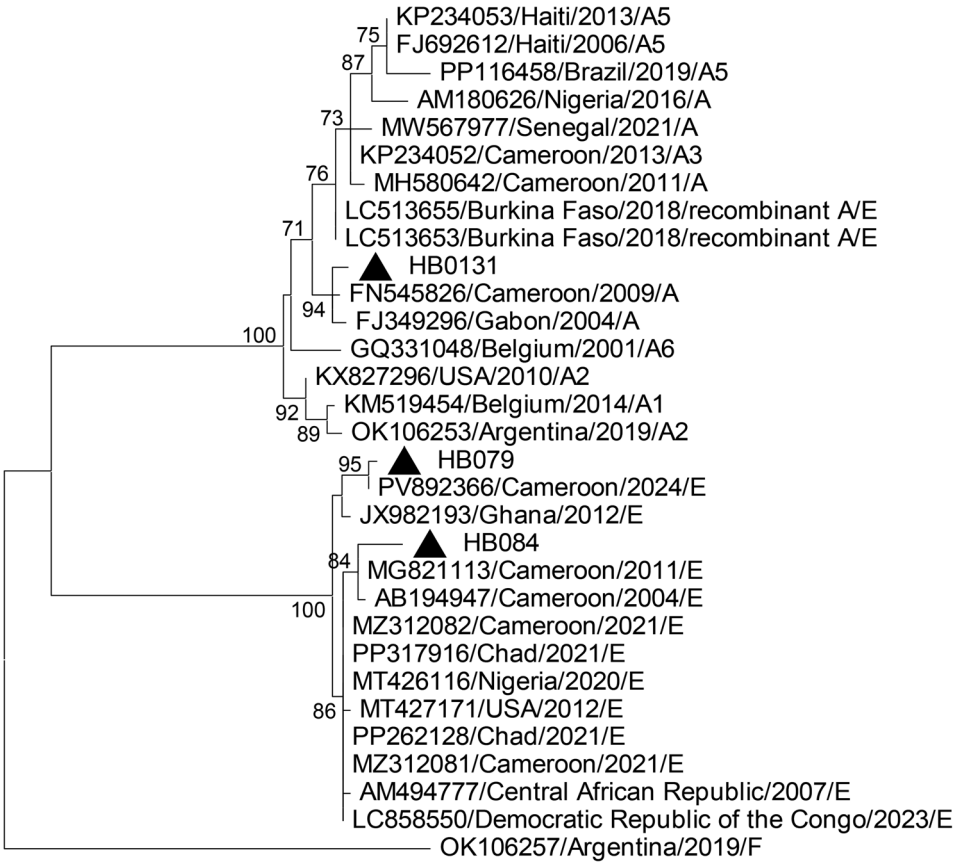


Fig. 1 Phylogenetic tree derived from fragment of the Pre-S region of HBV sequences. Only bootstrap values > 70% are presented. The strains identified in this study are indicated by triangles and lab numbers. Phylogenetic analysis was performed using the Maximum Likelihood method and Kimura 2-parameter model, using MEGA software (Version 7). For each reference sequence (n = 28) obtained from GenBank, accession number, country of origin, year of collection and genotype (sub-genotype for those available) are provided

In developed countries, systematic viral load screening nearly eliminates OBI transmission risk [40]. In low- and middle-income countries, access to resources and advanced technologies, such as PCR, is limited. As a result, the WHO's goal of eliminating hepatitis B by 2030 appears very difficult to achieve.

Using Fisher's exact test with a significance threshold of $p = 0.05$, we identified the number of prior donations as a significant factor negatively associated with OBI ($p = 0.021$). This suggests that repeated screening of regular donors significantly could reduce the risk of transfusion-transmitted OBI. First-time donors had a higher HBV DNA positivity rate (5.6%) compared to repeat donors (0.8%), with an odds ratio of 0.13 [0,0–0,80] indicating an 86% lower chance of HBV DNA positivity in repeat donors.

Phylogenetic analysis revealed the co-circulation of HBV genotypes A and E, confirming the results of the recent study conducted in Southwest region of Cameroon [21]. These results also align with findings in Burkina Faso, where genotypes E and A co-circulate [41]. In Nigeria, only genotype A was detected among blood donors [42]. The non-detection of OBI may be attributed to viral variants with surface protein mutations that render HBsAg unrecognizable by standard serological tests [43]. Sequenced samples in this study showed low viral loads (< 30 IU/mL), suggesting strong immune suppression of viral replication and S gene expression, driven by immune responses targeting cccDNA elimination [44].

Unlike cases of OBI detected in people living with HIV or chronic liver disease, as described in previous studies, this study highlighted silent HBV infection among a group of volunteer blood donors in a context of reduced blood donations. In light of these findings, greater attention should be paid to blood testing for HBV screening in donor selection, as this contributes to increased rates of transfusion-related HBV transmission. The current findings provide further evidence of the risk of OBI transmission by apparently healthy donors, which must be adequately addressed. Blood screening policies have improved in recent years in Cameroon, but not to the point of detecting the HBV genome in suspected OBI cases. Indeed, lack of resources remains the main obstacle to the implementation of systematic NAT screening

for these cases. However, total elimination requires a strengthened policy for the elimination of OBI, through systematic screening of the viral load of all sources, including the healthy population and blood donors.

Our study has weaknesses or limitations. The selection of donors based on HBsAg negative and anti-HBc positive status excludes the rarer cases of anti-HBs alone and OBI seronegative status. According to the literature, true OBIs with a viral load above 1,000 IU/mL are very rare. In our study, two samples, 222 and 281, which had viral loads of 9,437,234 IU/mL and 5,354,835 IU/mL, respectively, were retested using another quantitative molecular technique (Abbott), and the results were almost identical. They were then tested serologically using another ELISA test (Roche), and the results for HBsAg were still negative. It is also surprising that the samples with the highest viral loads (172, 222, 243, and 281) did not yield sequences. Our attempts (increasing the volume of serum extracted or the volume of eluate and/or using degenerate primers for nested PCR) to obtain sequences from these samples were unsuccessful. We believe that mutations may be responsible for these inconsistencies and that the use of next-generation sequencing in a future larger study would help answer the questions that remain unresolved. With regard to risk factors, it would have been relevant to include in our study a more in-depth analysis of confounders, such as health-seeking behavior and screening bias which could explain the observed link between repeated blood donations and reduced risk of OBI.

Conclusion

Our study highlights that only sensitive HBV DNA screening would improve blood safety since anti-HBc screening would exclude 60% of donors in a situation of blood shortage. However, the cost of such testing, even in the low yield pools of 6 as practice in some countries, is likely prohibitive in Cameroon. However, the high frequency of prior infection of patients would limit transmission except in young children who should be vaccinated. In present Cameroon situation, systematic, compulsory, vaccination would be the best affordable prevention of OBI transfusion transmission.

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Author contributions
Conceptualization: AKBC, TFT, SE, RN; Methodology: RN, AFM, TFT, SE, AKBC; Validation: RN, AFM, NAT, TNNN, AKBC; Investigation: AKBC; Writing-original draft preparation: AKBC; Writing-review and editing: TFT, SE, AFM, RN; Supervision: TFT, RN; Project Administration: SE, TFT, NAT, RN; Funding: RN. All authors have read and agreed to the published version of the manuscript.

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Data availability
The HBV strains identified in this study are available in GenBank (GenBank accession numbers: PX313721-PX313723).

Declarations

Ethics approval and consent to participate
The study protocol was reviewed and approved by the Regional Ethics Committee for Human Health Research of the Center (CRERSH/C) (Number: 000734/CRERSHC/2025) and adhered to the ethical guidelines outlined in the 1975 Declaration of Helsinki. Written informed consent was obtained from each participant before enrollment, in accordance with ethical guidelines, ensuring that any personal data or identifiable information will be appropriately anonymized to protect individual privacy.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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